AN/SSQ-53F DIFAR

- **Mircoprocessor-Controlled Full-Function Selector**
  - **Electronic Function Select before launch**
  - **Command Function Select after launch**
- **Multichannel RF Capability**
- **Multiple Depth and Life Selection Capability**
- **DIFAR or Constant Shallow Omni, Calibrated Omni Sensors**
- **Launchable from Existing ASW Platforms**

The AN/SSQ-53F DIFAR Sonobuoy is the latest Directional LOFAR sonobuoy being produced for the US Navy. Basic capabilities include four-depth operation (90, 200, 400 and 1000 feet), five life selections (.5, 1.0, 2.0, 4.0 and 8.0 hours), choice of DIFAR or Constant Shallow Omni Sensor, Calibrated Omni, choice of DIFAR AGC ON/OFF, and the capability to set any one of ninety-six VHF Transmitter channels. These channels may be set via Electronic Function Selection (EFS) prior to launch or after deployment via Command Function Select (CFS). After deployment the following selections can be changed via the Command Function Selection (CFS): Life, RF Channel, Sensor type, and DIFAR AGC ON/OFF.

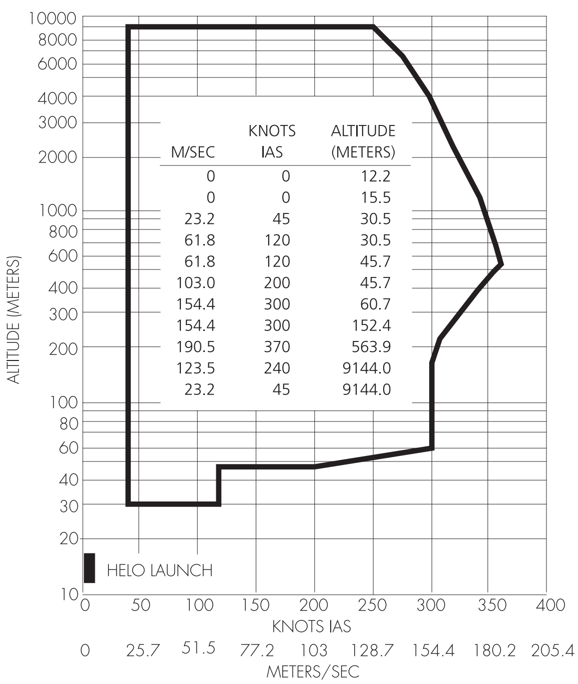
The AN/SSQ-53F can be air launched at airspeeds up to 370 KIAS and altitudes up to 30,000 feet or can be launched from helicopters as low as 40 feet with zero forward velocity.

Passive Sonobuoy

Sparton Government Business Division
5612 Johnson Lake Road - DeLeon Springs, FL 32130 - government@sparton.com
Phone 386-985-4631 - Toll Free 800-952-7589 - FAX 386-985-9855

Type ………………………...…………	**Passive, directional, low frequency**
Size ……………………………….....….	**A**
Weight …………………………………..	**22 pounds (10 kilograms)**
Launch …………………………....……	**40 to 30,000 Feet (12 to 9144 Meters)**
Operational Depth ……………………..	**at 0 to 370 KIAS**
	DIFAR & CO 90, 200, 400, and 1000 Feet (27, 61, 122, and 305 Meters)
	CSO ………... 45 Feet (13.7 Meters)
Operational Life ……………………....	**5, 1.0, 2.0, 4.0, and 8.0 Hours (EFS or CFS selectable)**
RF Channels …………………………….	**96 (136 to 173.5 MHz) (EFS or CFS selectable)**
RF Power ………………………………..	**1.0 Watt**
Scuttle …………………………………..	**8 Hours**
Audio Frequencies …………………....	**5 to 2400 Hz. DIFAR**
	30 to 5000 Hz. CSO
	5 to 20 kHz CO

Technical Specifications



SPECIFICATIONS

PHYSICAL CHARACTERISTICS

Weight .... *8.2 kg (18 lbs.)*

Sonobuoy Launch Container (SLC) *LAU–126/A*

ENGINEERING DATA

Transmitter Power Output .... *1 Watt Minimum*

Operating Frequency .... *99 Selectable Channels (136.000 to 173.500 MHz)*

Transmitter Frequency Stability .... *± 25 kHz*

Frequency Modulation .... *0 to 105 kHz Deviation*

Multiplexed Composite consisting of:

- ***Omni*** .... *Direct Summation*
- ***Directional Data (2 Channels)*** .... *Suppressed Carrier Quadrature Modulated on Subcarriers*
- ***Frequency Reference Pilot*** .... *Direct Summation*
- ***Phase Reference Pilot*** .... *Direct Summation*
- ***Harmonic Distortion*** .... *5% Maximum*
- ***Modulation Asymmetry*** .... *20% @ 75 kHz Deviation*

Operating Life .... *Selectable 0.5, 1, 2, 4 or 8 Hours*

Operating Depth .... *Selectable 30m, 60m, 120m or 300m (100', 200', 400' or 1000')*

Hydrophone .... *Multi-Channel Directional (Piezoelectric Ceramic)*

Shelf Life .... *5 Years in Sealed Container*

Unpacked Storage Life (Minimum) .... *90 Days*

Selections (Life, Depth and RF Channel) .... *Electronic Function Select*

UnderSea Sensor Systems Inc.
4578 East Park 30 Drive
Columbia City, IN
USA 46725-8869

Tel: 219.248.3500
Fax: 219.248.3510
email: mktg@undersea-sensors.com

Hermes Electronics Inc.
40 Atlantic Street
Dartmouth, Nova Scotia
Canada B2Y 4N2

Tel: 902.466.7491
Fax: 902.463.6098
email: mktg@hermes.ns.ca

04/00/AN/SSQ–53F
Printed in Canada, 2000